THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – I : Introduction & Basic Concepts

★ Note: Syllabus: System, boundary, surroundings, control volume, universe, types of systems, macroscopic & microscopic views, continuum, equilibrium, state, property, process, cycle, reversibility, quasi-static & irreversible processes, causes of irreversibility

1. What is Thermodynamics?

Thermodynamics is the science of energy – its storage, transformation (heat ⇄ work) and the properties of matter involved. The word comes from Greek: therme (heat) + dynamis (power).

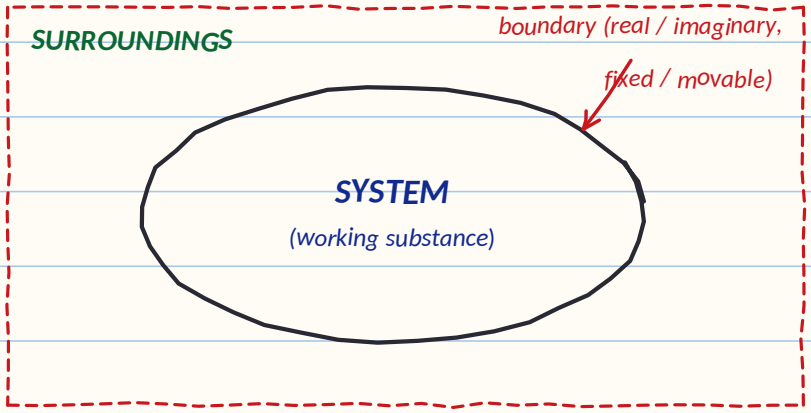
- Deals with energy interactions between a system and its surroundings, not with the rate (that is heat transfer).
- Based on experimental laws – Zeroth, First, Second and Third – which have never been violated.
- Applications: IC engines, steam & gas turbines, refrigerators, air-conditioners, compressors, rockets, power plants.
- Two approaches: macroscopic (classical) and microscopic (statistical).

2. System, Boundary, Surroundings, Universe

- System : a definite quantity of matter or a region in space chosen for study.
- Surroundings : everything outside the system that can interact with it.
- Boundary : real or imaginary surface separating system and surroundings. It may be fixed or movable. It has zero thickness and zero

volume.

- Universe : System + Surroundings.



Universe = System + Surroundings

Control volume : fixed region of space

Control surface : boundary of the CV

Boundary : thickness = 0, no volume

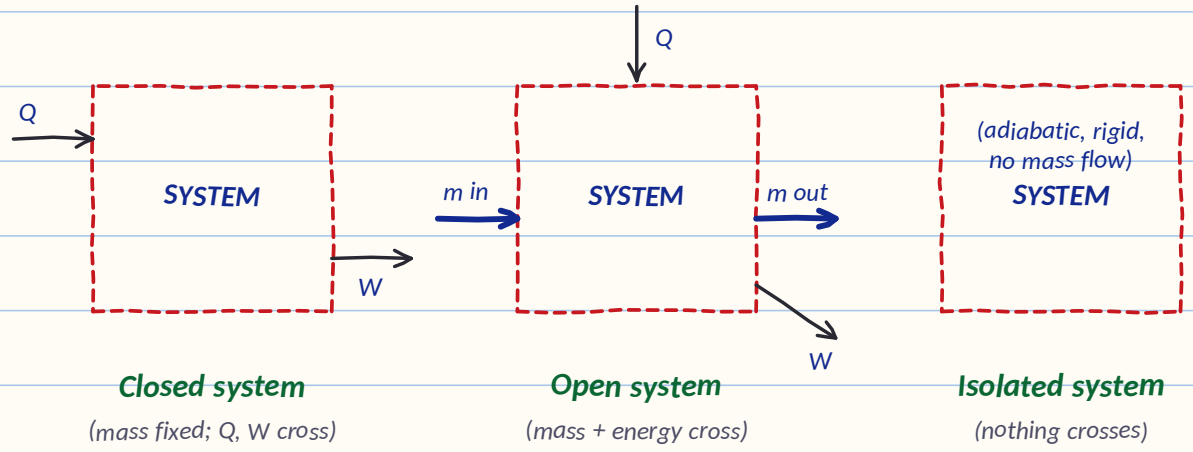
Types: closed, open, isolated

Control volume & control surface

A control volume is a fixed region in space (e.g. a turbine, nozzle, compressor) across which mass can flow. Its boundary is called the control surface. Used for open-system (flow) analysis.

3. Types of Systems

Type	Mass transfer	Energy transfer	Example
Closed (control mass)	No	Yes (Q, W)	Gas in piston-cylinder
Open (control volume)	Yes	Yes	Turbine, compressor, nozzle
Isolated	No	No	Thermos flask (ideal), universe



- Adiabatic system: no heat crosses the boundary ($Q = 0$). Rigid system: no work crosses ($W = 0$).
- Homogeneous system: single phase. Heterogeneous system: more than one phase (e.g. ice + water).

4. Macroscopic and Microscopic Viewpoints

Macroscopic (classical)	Microscopic (statistical)
Considers the matter as a whole	Considers molecules individually
No assumption about structure	Needs model of molecular structure
Few properties: P, V, T measured	Many variables: velocity, position of each
Simple, uses senses/instruments	Complex, uses statistics & probability
Properties are averages of molecules	Behaviour of each molecule is studied

★ Note: In this course we use the macroscopic approach. It works because a system consists of a very large number of molecules, so the average behaviour is steady and measurable.

5. Concept of Continuum

Matter is made of discrete molecules with empty space between. Classical thermodynamics ignores this and treats matter as a continuous medium (continuum) so that properties such as density, pressure and temperature vary smoothly from point to point.

- Density at a point: $\rho = \lim (\delta m/\delta V)$ as $\delta V \to \delta V'$, where $\delta V'$ is the smallest volume that still contains enough molecules to give a statistical average.

$$\rho = \lim \delta m / \delta V \quad (\delta V \to \delta V')$$

- Valid when the mean free path of molecules $\ll$ characteristic dimension of the system.
- Fails in highly rarefied gases (very high vacuum, upper atmosphere) – there the Knudsen number is large.

6. Thermodynamic Property, State, Path

- Property : any observable characteristic of a system (P, V, T, U, H, S ...). It depends only on the state, not on how the state was reached.
- Intensive properties: independent of mass (P, T, density, specific volume). Extensive properties: depend on mass (V, U, H, S, m).
- Specific property = extensive / mass, e.g. $v = V/m$, $u = U/m$. Specific extensive property behaves like an intensive one.
- State : the condition of a system described by its properties. Two independent properties fix the state of a simple compressible pure substance.
- Path : the series of states passed through. Process : change of state from one equilibrium state to another. Cycle : a series of processes after which

the system returns to its initial state.

★ Note: Heat and work are NOT properties – they are path functions. dQ, dW are inexact differentials; dU, dH, dS are exact (point functions).

7. Thermodynamic Equilibrium

A system is in thermodynamic equilibrium when its properties do not change with time and there is no spontaneous change when it is isolated. It requires three types of equilibrium to hold together:

→ Mechanical equilibrium – no unbalanced force; pressure uniform (no pressure gradient).

→ Thermal equilibrium – no temperature difference; temperature uniform.

→ Chemical equilibrium – no chemical reaction, no mass transfer by concentration difference.

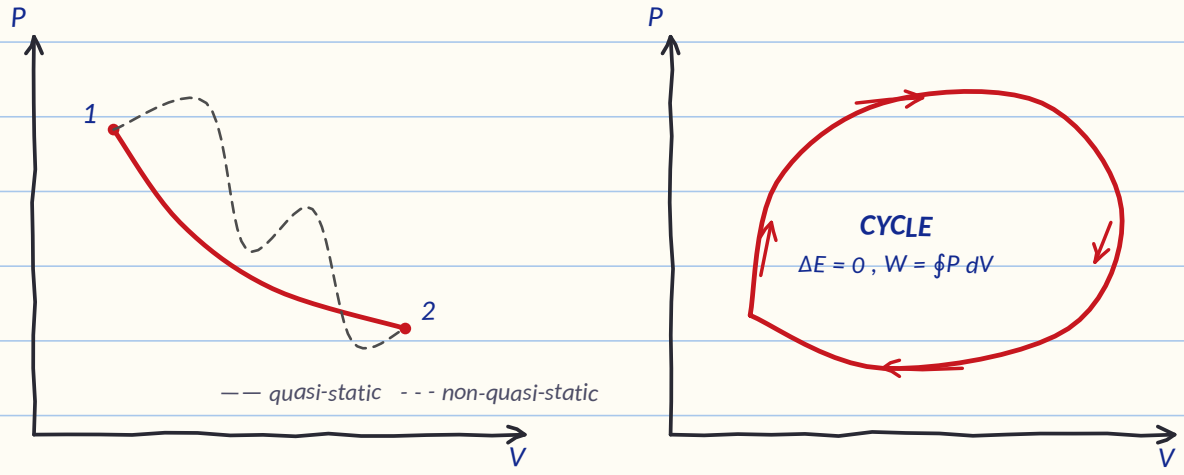
- Only equilibrium states can be represented as points on a property diagram (P-V, T-S etc.).
- Phase equilibrium is also included when more than one phase is present.

8. Processes and Cycles

Common processes on a P-V diagram (n = polytropic index, $PV^n = C$ written as $PV^n = C$):

Process	Constant	Index n	Work 1→2
Isobaric	P	0	P(V2 – V1)
Isochoric	V	∞	0
Isothermal	T (PV)	1	P1V1 ln(V2/V1)
Adiabatic	Q = 0	γ	(P1V1 – P2V2)/(γ–1)
Polytropic	PV^n	n	(P1V1 – P2V2)/(n–1)

Cycle: process 1-2-3-4-1 returns to initial state, so ΔE = 0 and net work = net heat = area enclosed. Clockwise on P-V = power cycle (engine); anticlockwise = refrigeration / heat-pump cycle.



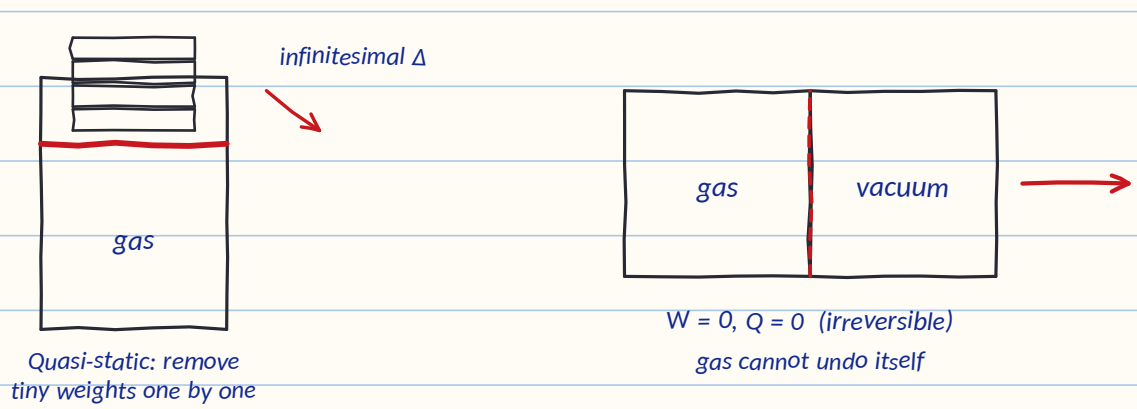
9. Quasi-static (Quasi-equilibrium) Process

A process that proceeds so slowly that the system passes through a continuous series of equilibrium states – each intermediate state is infinitesimally close to equilibrium.

- Example: gas in a cylinder with a piston carrying many small weights – remove one tiny weight at a time and let the gas settle.
- Can be drawn as a continuous line on a property diagram; a non-quasi-static process can only be shown with dotted line joining end

points.

- It is an idealisation. Real processes are fast, but many (like slow engine strokes) approximate it. It gives the maximum work output for expansion and minimum work input for compression.



10. Reversible and Irreversible Processes

Reversible process

A process which can be reversed so that the system and surroundings both return exactly to their initial states, leaving no trace in the universe.

- Must be quasi-static AND free from friction, free expansion, finite temperature-difference heat transfer, mixing etc.
- Every reversible process is quasi-static, but not every quasi-static process is reversible (e.g. quasi-static with friction).
- Gives maximum efficiency; used as the ideal standard (Carnot cycle).

Irreversible process

A process that cannot be reversed without leaving some change in the system or surroundings. All natural / real processes are irreversible.

- Examples: free expansion, heat flow from hot to cold body, mixing of two gases, combustion, flow through a valve (throttling), friction, electrical

resistance heating.

11. Causes of Irreversibility

Cause	Explanation / example
Friction	Mechanical (solid-solid), fluid friction – work turned to heat
Unrestrained expansion	Free expansion of gas into vacuum
Heat transfer through finite ΔT	Heat from a hot to a cold body
Mixing	Two different fluids / gases mixing
Chemical reactions	Combustion, diffusion, dissolving
Electrical effects	I^2R loss, hysteresis, magnetization losses
Inelastic deformation	Plastic deformation of solids
Non-quasi-static process	Finite pressure / temperature gradients

★ Note: Internal irreversibility occurs within the system (friction, turbulence, free expansion). External irreversibility occurs at the boundary (heat transfer through finite temperature difference).

12. Units, Pressure and Temperature

- SI base units: length m, mass kg, time s, temperature K, amount of substance mol.
- Force: 1 N = 1 kg·m/s². Pressure: 1 Pa = 1 N/m². Energy: 1 J = 1 N·m. Power: 1 W = 1 J/s.

$1\ bar = 10^5\ Pa = 100\ kPa \quad 1\ atm = 1.01325\ bar = 760\ mm\ Hg$

$1\ kWh = 3600\ kJ \quad 1\ kJ/kg = 1000\ m^2/s^2$

Types of pressure

- Absolute pressure = pressure measured from perfect vacuum (zero). Used in all thermodynamic equations.

- Gauge pressure = pressure above atmosphere; vacuum pressure = pressure below atmosphere.

$$P_{abs} = P_{atm} + P_{gauge}$$

$$P_{abs} = P_{atm} - P_{vacuum}$$

- Manometer / barometer: pressure of a fluid column $P = \rho g h$ (ρ of mercury = 13600 kg/m³).

Temperature scales

$$K = °C + 273.15 \qquad °F = 1.8\,°C + 32$$

$$°R = °F + 459.67 = 1.8\,K$$

- Zeroth law gives the basis of temperature measurement. Fixed point: triple point of water = 273.16 K (0.01 °C).
- Thermometers: liquid-in-glass (mercury, alcohol), constant-volume gas, resistance (RTD), thermocouple, optical pyrometer.

13. Equation of State and State Postulate

The relation among P, V and T of a substance is its equation of state. For an ideal (perfect) gas:

$$P V = m R T \quad \text{or} \quad P v = R T \quad \text{or} \quad P V = n R_u T$$

- $R = R_u/M$, universal gas constant R_u = 8.314 kJ/kmol·K. For air R = 0.287 kJ/kg·K.
- Gas laws: Boyle (T const: PV = C), Charles (P const: V/T = C), Gay-Lussac (V const: P/T = C), Avogadro (equal volumes, same P & T, same number of molecules).
- Real gases deviate at high pressure / low temperature: van der Waals equation $(P + a/v^2)(v - b) = RT$.

★ Note: State postulate: the state of a simple compressible system is completely fixed by two independent intensive properties. Example: (P, T) or (P, v) or (T, v).

Point function vs Path function

Point function	Path function
Depends only on end states	Depends on the path followed
Exact differential (dU, dP, dV)	Inexact differential (δQ, δW)
Cyclic integral $\oint dX = 0$	Cyclic integral $\oint \delta Q$, $\oint \delta W \neq 0$
Examples: P, T, V, U, H, S	Examples: heat, work

14. Solved Numerical Problems

Problem 1 : Absolute pressure

A pressure gauge on a tank reads 2.5 bar. Barometer reads 760 mm Hg. Find absolute pressure.

$$P_{abs} = 2.5 + 1.01325 = 3.513 \text{ bar}$$

Problem 2 : Vacuum gauge

A condenser vacuum gauge reads 400 mm Hg when barometer reads 760 mm Hg. Find the absolute pressure.

$$P_{abs} = 760 - 400 = 360 \text{ mm Hg} = (360/760)(1.01325) = 0.48 \text{ bar}$$

Problem 3 : Manometer

A U-tube mercury manometer shows a difference of 0.2 m. Find the gauge pressure.

$$\Delta P = \rho g h = 13600 \times 9.81 \times 0.2 = 26\,683 \text{ Pa} \approx 26.7 \text{ kPa}$$

Problem 4 : Isothermal expansion work

1 kg of air expands isothermally at 300 K from 5 bar to 1 bar. Find the work done ($R = 0.287$ kJ/kg·K).

$$W = m\,R\,T\,\ln(P_1/P_2) = 1 \times 0.287 \times 300 \times \ln 5$$

$$W = 86.1 \times 1.609 = 138.6\ kJ$$

Problem 5 : Polytropic expansion

A gas at 4 bar, 0.02 m^3 expands as $PV^{1.3} = C$ to 0.06 m^3. Find final pressure and work.

$$P_2 = P_1(V_1/V_2)^{1.3} = 4\ (1/3)^{1.3} = 0.959\ bar$$

$$W = (P_1V_1 - P_2V_2)/(n-1) = (8000 - 5756)/0.3 = 7481\ J \approx 7.48\ kJ$$

Problem 6 : Net work of a rectangular cycle

Cycle: 1-2 isobaric at 2 bar, V from 0.1 to 0.3 m^3; 2-3 isochoric to 1 bar; 3-4 isobaric back to 0.1 m^3; 4-1 isochoric.

$$W_{12} = 2\times10^5(0.2) = +40\ kJ\ ;\quad W_{34} = 1\times10^5(-0.2) = -20\ kJ$$

$$W_{net} = 40 - 20 = 20\ kJ\ \ (= \text{enclosed area})$$

15. Quick Revision – Important Points

- System + Surroundings = Universe; boundary has zero thickness.
- Closed: mass fixed. Open: mass crosses. Isolated: neither mass nor energy.
- Property = point function (exact differential); heat & work = path functions.
- Equilibrium = mechanical + thermal + chemical (+ phase).
- Quasi-static ⇒ infinite slowness; reversible ⇒ quasi-static + no dissipation.
- Cycle: $\oint dP = \oint dV = \oint dT = 0$ for any property; $\oint dQ = \oint dW$.

Frequently asked questions

→ Differentiate between closed and open systems with examples.

→ Explain quasi-static process with a neat sketch. How is it different from

reversible?

→ What are the causes of irreversibility? Explain with examples.

→ Explain thermodynamic equilibrium and its types.

→ State the concept of continuum. When does it fail?

16. Short Answer Questions (Exam Ready)

→ Define thermodynamics. Science of energy and its transformations between heat and work, and the properties of the matter involved.

→ What is a thermodynamic property? A point function describing the state, independent of the path (P, T, V, U, H, S).

→ Intensive vs extensive? Intensive – independent of mass (P, T); extensive – proportional to mass (V, U, H).

→ What is a cycle? A series of processes that returns the system to its initial state; $\oint dX = 0$ for properties.

→ Why is free expansion irreversible? The gas cannot return to the original container without external work, and no work is obtained during expansion.

→ State the zeroth requirement for equilibrium. Mechanical, thermal and chemical equilibrium must exist together.

→ Is every quasi-static process reversible? No. A quasi-static process with friction is not reversible.

→ Examples of open systems? Boiler, turbine, nozzle, compressor, pump, heat exchanger.

~ End of Unit I ~